

# Evaluating and Predicting CardioVascular Diseases using UCI Heart Dataset

Surafel Melese Tariku

May 2025

## Introduction

The UCI Heart Disease dataset is a widely used benchmark for developing and evaluating machine learning models in medical diagnostics. It contains 76 attributes, though most analyses focus on a subset of 14 features. These include clinical and demographic factors like age, cholesterol level, and resting blood pressure. The target variable indicates the presence of heart disease, making the dataset suitable for binary classification tasks in predictive health analytics.

## Background

Cardiovascular diseases (CVDs) are a leading cause of global mortality. While traditional risk models exist, they often lack accuracy for complex cases. Deep learning offers superior prediction but operates as a "black box." Explainable AI (XAI) is crucial for transparency, fostering trust in clinical settings. Integrating XAI with deep learning for CVD not only enhances accuracy but also validates predictions against known risk factors, building a robust and interpretable framework.

## Project Objectives

The primary objectives of this project are threefold:

1. To develop an advanced deep learning model capable of accurately predicting cardiovascular disease (CVD) utilizing comprehensive patient data.
2. To integrate state-of-the-art Explainable AI (XAI) techniques to elucidate the key features driving the model's predictions, thereby enhancing transparency and trust.
3. To rigorously compare the insights gleaned from the XAI explanations with the findings from an independent exploratory data analysis (EDA), ensuring consistency in the identified cardiovascular risk factors.

## Methodology

The initial step in data preparation involved robust **outlier detection and removal** to mitigate their adverse impact on model performance. Subsequently, the entire dataset underwent **normalization**, scaling all features to a consistent range to ensure fair contribution during model training.

**For non-deep learning models, the preprocessed data was partitioned into a training set (70%) and a test set (30%). For the deep learning models, the data was split into three distinct sets: a training set (70%), a validation set (15%), and a test set (15%).** Crucially, **categorical variables were not one-hot encoded** as the dataset does not contain such features.

## Gaussian Naive Bayes Model

As an initial baseline for comparison, a **Gaussian Naive Bayes (GNB) model** has been implemented. This probabilistic classifier assumes that features are conditionally independent given the class label, and that the likelihood of the features is Gaussian (normally distributed). It is important to note that our exploratory data analysis (EDA) on the UCI heart dataset indicated that several features are not independent, which violates a core assumption of the GNB model.

A significant advantage of the GNB model for this initial phase is that it **does not have hyperparameters that require tuning**. This characteristic simplifies the implementation and evaluation process, allowing for a quick and unoptimized baseline performance measurement before proceeding with more complex deep learning architectures that typically necessitate extensive hyperparameter optimization.

## Gaussian Naive Bayes Model Results

The performance of the Gaussian Naive Bayes (GNB) model, serving as our initial baseline, is summarized in the table below.

Table 1: Gaussian Naive Bayes Model Performance

| Metric   | Training Set | Test Set |
|----------|--------------|----------|
| Accuracy | 0.866        | 0.852    |
| F1 Score | 0.840        | 0.719    |

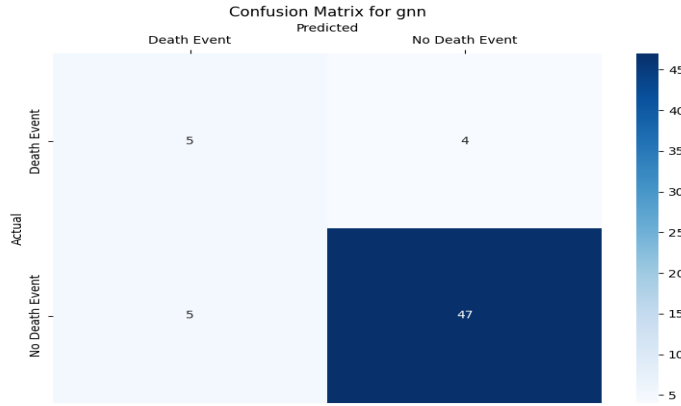


Figure 1: Confusion Matrix for Gaussian Naive Bayes Model on the Test Set.

## Explainable AI with SHAP for Gaussian Naive Bayes

This SHAP summary plot explains the impact of each feature on the model's predictions. `time` and `ejection_fraction` are the most influential, with longer `time` reducing risk and lower `ejection_fraction` increasing it. Features like `serum_creatinine` and `age` also significantly affect predictions.

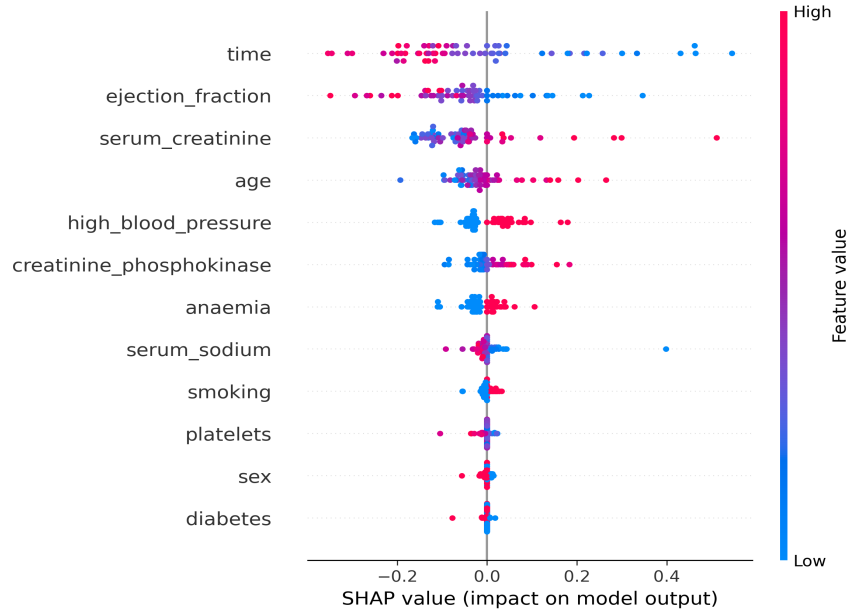


Figure 2: SHAP summary plot for our Guassian Naive bias Model.

### Explainable AI with LIME for Gaussian Naive Bias

Unlike SHAP, which can offer both global and local explanations, LIME focuses specifically on providing **local explanations**, making individual predictions interpretable. This complements the global feature importance provided by SHAP, offering a comprehensive view of the model's decision-making process.

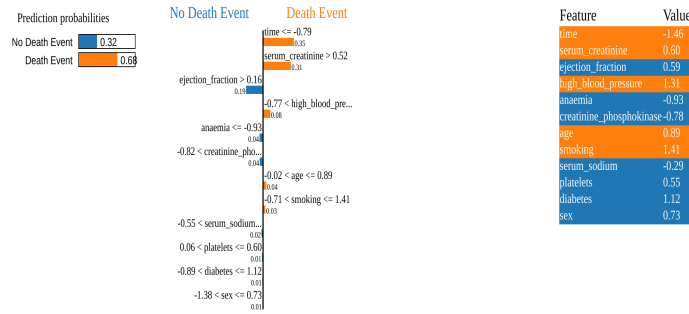


Figure 3: LIME summary for our Guassian Naive bias Model for a sample image

| Feature                  | EDA (Global Insight)                             | LIME (Local Explanation)                   |
|--------------------------|--------------------------------------------------|--------------------------------------------|
| Age                      | Older patients show higher risk of heart disease | Within normal range; small contribution    |
| Cholesterol              | Higher levels linked to heart disease            | Not shown as significant for this instance |
| Max Heart Rate (thalach) | Lower rates correlate with disease               | Not considered in the model                |
| Oldpeak                  | Higher values indicate greater risk              | Not considered in the model                |
| Sex                      | Males more likely to have heart disease          | Present; small impact                      |
| Time (follow-up)         | Not analyzed                                     | Very low; strongly increases death risk    |
| Serum Creatinine         | Not emphasized globally                          | High; strongly increases risk              |
| Ejection Fraction        | Not emphasized globally                          | Slightly high; mildly reduces risk         |
| High Blood Pressure      | Mild global association                          | Present; increases predicted risk          |
| Anaemia                  | Not highlighted globally                         | Not anaemic; slightly protective           |
| Serum Sodium             | Not emphasized globally                          | Low; mildly increases risk                 |
| Platelets                | Not emphasized globally                          | Slightly high; minor protective impact     |
| Diabetes                 | Not strongly linked in EDA                       | Present; very minor effect                 |
| Smoking                  | Not emphasized globally                          | Present; small influence on risk           |

Table 2: Comparison of Global EDA Insights and Local LIME Explanation for Heart Disease Prediction

### Logistic Regression Classifier

Following the baseline Gaussian Naive Bayes model, a **Logistic Regression classifier** will be implemented. Logistic Regression, while a linear model, is a widely used and robust algorithm for binary classification problems, serving as another strong benchmark before deep learning approaches. We will utilize the implementation available in `sklearn.linear_model`.

The performance of the Logistic Regression model will be optimized through the tuning of key hyperparameters:

- **C**: This inverse of regularization strength; smaller values specify stronger regularization. We will explore values of **0.1**, **0.6**, and **1.1**.
- **penalty**: Specifies the norm of the penalty.
- **solver**: Algorithm to use in the optimization problem.

This hyperparameter tuning will be conducted to identify the optimal configuration for predicting cardiovascular disease.

| Solver    | Penalty                  |
|-----------|--------------------------|
| lbfgs     | l2, None                 |
| newton-cg | l2, None                 |
| sag       | l2, None                 |
| saga      | elasticnet, l1, l2, None |

Table 3: Solvers and their corresponding penalty options in Logistic Regression

### Logistic Regression Classifier Results

The results of the hyperparameter tuning for the Logistic Regression classifier are presented in Table 4. This table details the performance metrics (Accuracy and F1 Score) on both the training and test sets for various combinations of the **C** (regularization strength), **penalty**, and **solver** hyperparameters.

### Optimal Logistic Regression Model Selection

Based on the hyperparameter tuning results presented in Table 4, the optimal configuration for the Logistic Regression model was identified. The best-performing model achieved the highest F1 Score on the test set, demonstrating a strong balance between precision and recall on unseen data. This optimal model was achieved with the following hyperparameters:

- **C = 0.1**

Table 4: Logistic Regression Model Performance Across Hyperparameter Settings

| C   | Penalty    | Solver    | Acc. Train (%) | F1 Train | Acc. Test (%) | F1 Test |
|-----|------------|-----------|----------------|----------|---------------|---------|
| 0.1 | l2         | lbfgs     | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.1 | l2         | newton-cg | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.1 | l2         | sag       | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.1 | l2         | saga      | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.1 | elasticnet | saga      | 83.8           | 0.818    | 80.3          | 0.707   |
| 0.1 | l1         | saga      | 81.7           | 0.793    | 80.3          | 0.707   |
| 0.6 | l2         | lbfgs     | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.6 | l2         | newton-cg | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.6 | l2         | sag       | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.6 | l2         | saga      | 83.8           | 0.816    | 75.4          | 0.643   |
| 0.6 | elasticnet | saga      | 83.1           | 0.807    | 78.7          | 0.673   |
| 0.6 | l1         | saga      | 82.4           | 0.800    | 73.8          | 0.629   |
| 1.1 | l2         | lbfgs     | 82.4           | 0.802    | 75.4          | 0.643   |
| 1.1 | l2         | newton-cg | 82.4           | 0.802    | 75.4          | 0.643   |
| 1.1 | l2         | sag       | 82.4           | 80.2     | 75.4          | 0.643   |
| 1.1 | l2         | saga      | 82.4           | 0.802    | 75.4          | 0.643   |
| 1.1 | elasticnet | saga      | 83.1           | 0.809    | 77.0          | 0.658   |
| 1.1 | l1         | saga      | 83.1           | 0.807    | 77.0          | 0.658   |

- **Penalty = elasticnet**
- **Solver = saga**

This specific combination yielded an **Accuracy of 80.3%** and an **F1 Score of 0.707** on the test set. The superior performance of the ‘elasticnet’ penalty with the ‘saga’ solver at a lower ‘C’ value suggests that a combination of L1 and L2 regularization is effective in preventing overfitting and enhancing generalization for this dataset, leading to a more robust predictive model for cardiovascular disease.



Figure 4: Confusion Matrix for Logistic Regression Model on the Test Set.

### Explainable AI with SHAP for the Optimal Logistic Regression Model

The SHAP summary plot reveals feature impacts on Logistic Regression predictions. **time**, **ejection fraction**, **serum creatinine**, and **age** are the most influential characteristics. Higher values of time, serum creatinine, and age generally increase the predicted risk (indicated by red dots shifting right on the SHAP value axis), while higher ejection fraction values tend to decrease the risk (red dots shifting left).

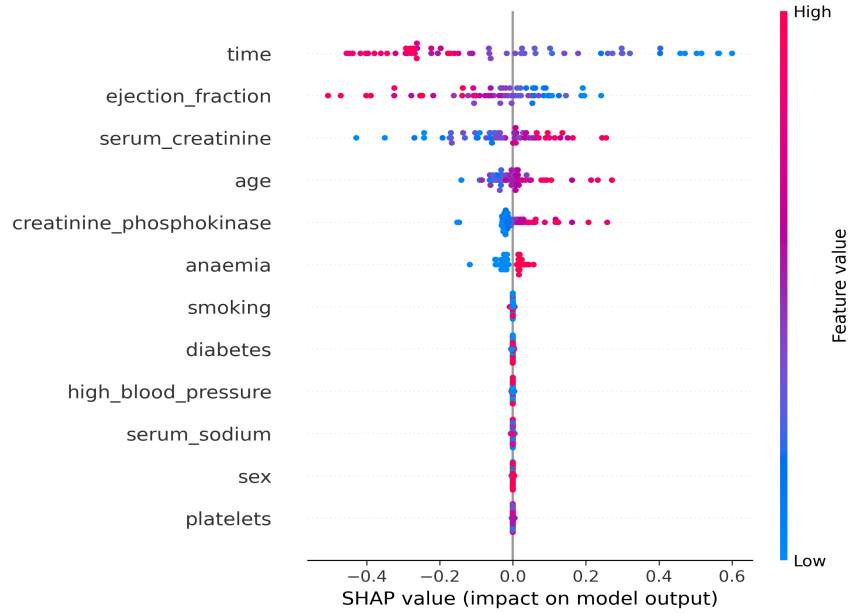


Figure 5: SHAP summary plot for the optimal Logistic Regression Model.

### Explainable AI with LIME for the Optimal Logistic Regression Model

Table 5: Comparison of Key Global EDA Insights and Local LIME Explanation for Heart Disease Prediction

| Feature           | EDA (Global Insight)                                                                          | LIME (Local Explanation for this instance)                                  |
|-------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Time              | Generally, longer follow-up duration is associated with better outcomes (less risk).          | Very low value (-1.46); strongly increases risk of death (+0.31).           |
| Ejection Fraction | A key indicator of heart pumping efficiency; lower values typically imply higher risk.        | High value (0.59); significantly decreases risk of death (-0.17).           |
| Serum Creatinine  | Indicator of kidney function; higher levels are generally associated with increased CVD risk. | High value (0.60); notably increases risk of death (+0.17).                 |
| Anaemia           | Often associated with increased cardiovascular risk and severity.                             | Not anaemic (low value of -0.93); slightly decreases risk of death (-0.02). |

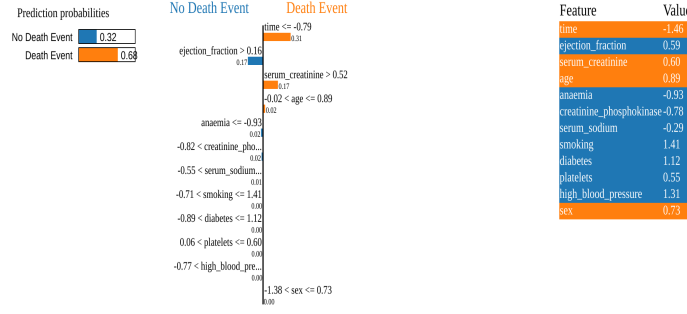


Figure 6: LIME summary for our Guassian Naive bias Model for a sample image

### Support Vector Classifier (SVC)

SVC is a powerful and versatile supervised machine learning algorithm used for classification. It works by finding an optimal hyperplane that best separates the data points into different classes in a high-dimensional feature space. It will be configured with a **probabilistic kernel** to enable the model to output probability estimates for classification. The primary hyperparameters to be tuned for the SVC model are:

- **C**: This regularization parameter controls the trade-off between achieving a low training error and a large margin. We will explore the following values for C: **0.2, 0.4, 0.6, and 0.8**.
- **gamma**: This parameter defines the influence of a single training example. The values considered for 'gamma' will be **'scale'** and **'auto'**.

### Support Vector Classifier (SVC) Results

The performance of the Support Vector Classifier (SVC) across various hyperparameter settings (C and gamma) is detailed in Table 6. The best combination has been identified to be C=0.8,

Table 6: SVC Model Performance Across Hyperparameter Settings

| C   | Gamma | Acc. Train (%) | F1 Train | Acc. Test (%) | F1 Test |
|-----|-------|----------------|----------|---------------|---------|
| 0.2 | auto  | 43.5           | 0.704    | 46.0          | 0.852   |
| 0.2 | scale | 43.5           | 0.704    | 46.0          | 0.852   |
| 0.4 | auto  | 79.1           | 0.852    | 61.3          | 0.852   |
| 0.4 | scale | 79.1           | 0.852    | 61.3          | 0.852   |
| 0.6 | auto  | 86.2           | 0.894    | 65.8          | 0.852   |
| 0.6 | scale | 86.2           | 0.894    | 65.8          | 0.852   |
| 0.8 | auto  | 91.2           | 0.930    | 65.8          | 0.852   |
| 0.8 | scale | 91.2           | 0.930    | 65.8          | 0.852   |

### Optimal Support Vector Classifier (SVC) Model Selection

Based on the hyperparameter tuning results presented in Table 6, the optimal Support Vector Classifier model is identified as the one configured with the following parameters:

- $C = 0.8$
- $\text{Gamma} = \text{auto}$

This model achieved a **training accuracy of 91.2%** and an **F1 Score of 0.930** on the training set. For the unseen test set, it demonstrated a **test accuracy of 65.8%** and an **F1 Score of 0.852**. This configuration represents the best balance of performance for the SVC model on this dataset.

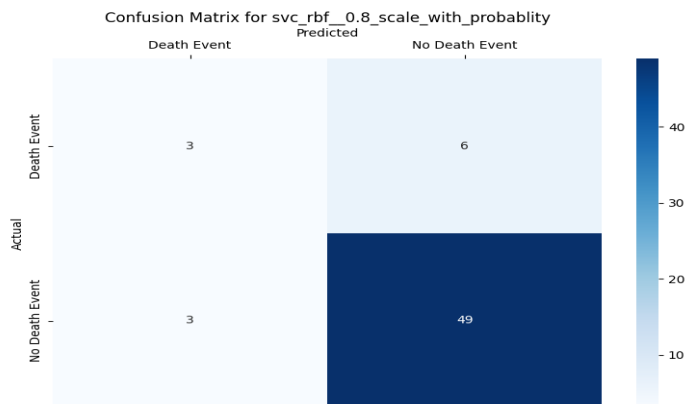


Figure 7: Confusion Matrix for Gaussian Naive Bayes Model on the Test Set.

### Explainable AI with SHAP for the Optimal Logistic Regression Model

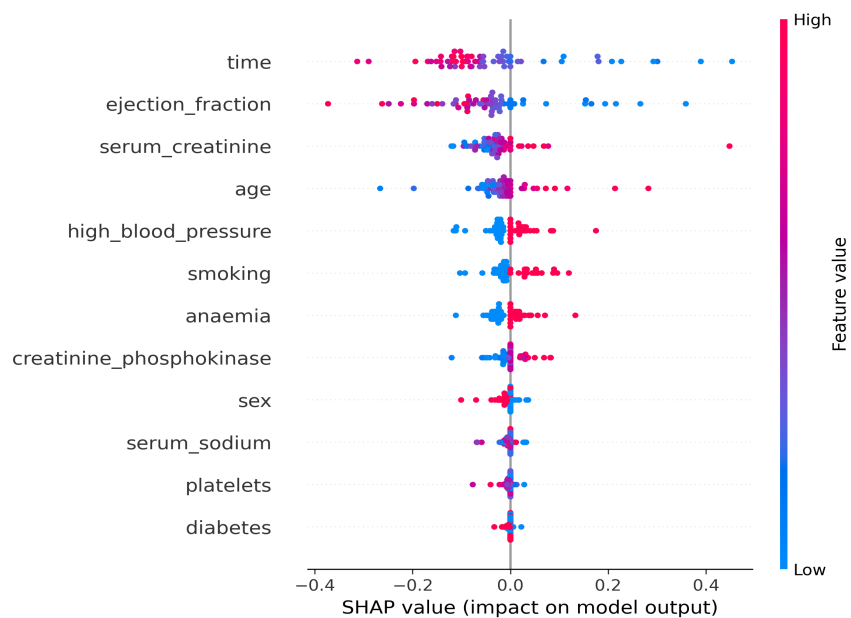


Figure 8: SHAP summary plot for Optimal SVC Model.



Table 7: Comparison of Key Global EDA Insights and Local SHAP Explanation for Optimal SVC Model

| Feature           | EDA (Global Insight)                                                                          | SHAP (Overall Impact for SVC Model)                                             |
|-------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Time              | Generally, longer follow-up duration is associated with better outcomes (less risk).          | Most influential feature; high values tend to increase the predicted risk.      |
| Ejection Fraction | A key indicator of heart pumping efficiency; lower values typically imply higher risk.        | Second most influential feature; low values strongly increase predicted risk.   |
| Serum Creatinine  | Indicator of kidney function; higher levels are generally associated with increased CVD risk. | Third most influential feature; high values notably increase predicted risk.    |
| Age               | Well-known primary risk factor for heart disease.                                             | Fourth most influential feature; high values generally increase predicted risk. |

### Explainable AI with LIME for the Optimal SVC Model

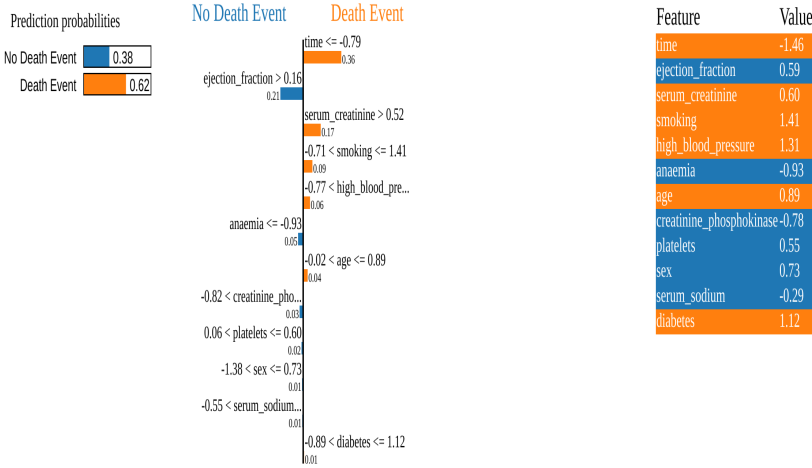


Figure 9: LIME summary for Optimal SVC Model for a sample image

Table 8: Comparison of Key Global EDA Insights and Local LIME Explanation for Optimal SVC Model

| Feature           | EDA (Global Insight)                                                                          | LIME (Local Explanation for this instance)                                      |
|-------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Time              | Generally, longer follow-up duration is associated with better outcomes (less risk).          | Very low value (-1.46); strongly increases the predicted risk of death (+0.36). |
| Ejection Fraction | A key indicator of heart pumping efficiency; lower values typically imply higher risk.        | High value (0.59); notably decreases the predicted risk of death (-0.21).       |
| Serum Creatinine  | Indicator of kidney function; higher levels are generally associated with increased CVD risk. | High value (0.60); considerably increases the predicted risk of death (+0.17).  |
| Smoking           | Major lifestyle risk factor for heart disease.                                                | High value (1.41); decreases the predicted risk of death (-0.09).               |

### K-Nearest Neighbors (KNN) Model

A **K-Nearest Neighbors (KNN)** classifier will be implemented as another non-deep learning model for cardiovascular disease prediction. Our number of neighbours is going to be 2.

### K-Nearest Neighbors (KNN) Model Results

The K-Nearest Neighbors (KNN) model, being a non-parametric algorithm, was evaluated without explicit hyperparameter tuning in this initial phase. Its performance is summarized in the table below.

Table 9: K-Nearest Neighbors (KNN) Model Performance

| Metric   | Training Set | Test Set |
|----------|--------------|----------|
| Accuracy | 81.7%        | 85.2%    |
| F1 Score | 0.725        | 0.658    |

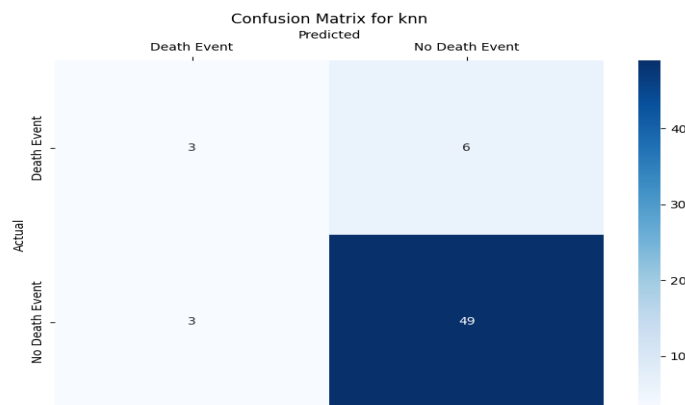


Figure 10: Confusion Matrix for KNN Model on the Test Set.

## Explainable AI with SHAP for KNN Model

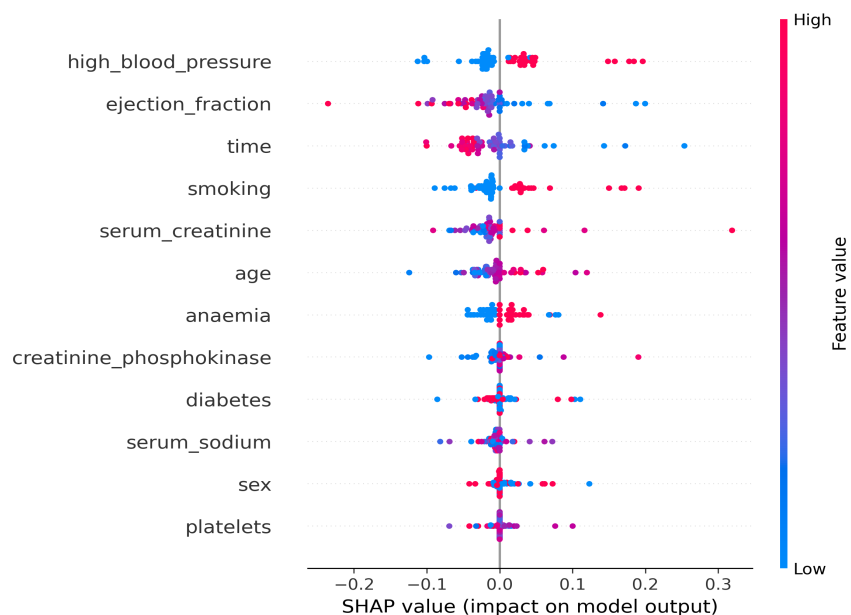


Figure 11: SHAP summary plot for our KNN Model.

Table 10: Comparison of Key Global EDA Insights and SHAP Explanation for KNN Model

| Feature             | EDA (Global Insight)                                                                   | SHAP (Overall Impact for KNN Model)                                                         |
|---------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| High Blood Pressure | Major risk factor for heart disease.                                                   | Most influential feature; high values tend to increase predicted risk, low values decrease. |
| Ejection Fraction   | A key indicator of heart pumping efficiency; lower values typically imply higher risk. | Second most influential; low values strongly increase predicted risk, high values decrease. |
| Time                | Generally, longer follow-up duration is associated with better outcomes (less risk).   | Third most influential; low values tend to increase predicted risk, high values decrease.   |
| Smoking             | Major lifestyle risk factor for heart disease.                                         | Fourth most influential; low values slightly decrease predicted risk, high values increase. |

## Explainable AI with LIME for the KNN Model

Table 11: Comparison of Key Global EDA Insights and Local LIME Explanation for KNN Model

| Feature           | EDA (Global Insight)                                                                   | LIME (Local Explanation for this instance)                                      |
|-------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Time              | Generally, longer follow-up duration is associated with better outcomes (less risk).   | Very low value (-1.46); strongly increases the predicted risk of death (+0.22). |
| Ejection Fraction | A key indicator of heart pumping efficiency; lower values typically imply higher risk. | High value (0.59); notably decreases the predicted risk of death (-0.21).       |
| Smoking           | Major lifestyle risk factor for heart disease.                                         | High value (1.41); notably increases the predicted risk of death (+0.11).       |
| Age               | Well-known primary risk factor for heart disease.                                      | High value (0.89); moderately increases the predicted risk of death (+0.06).    |

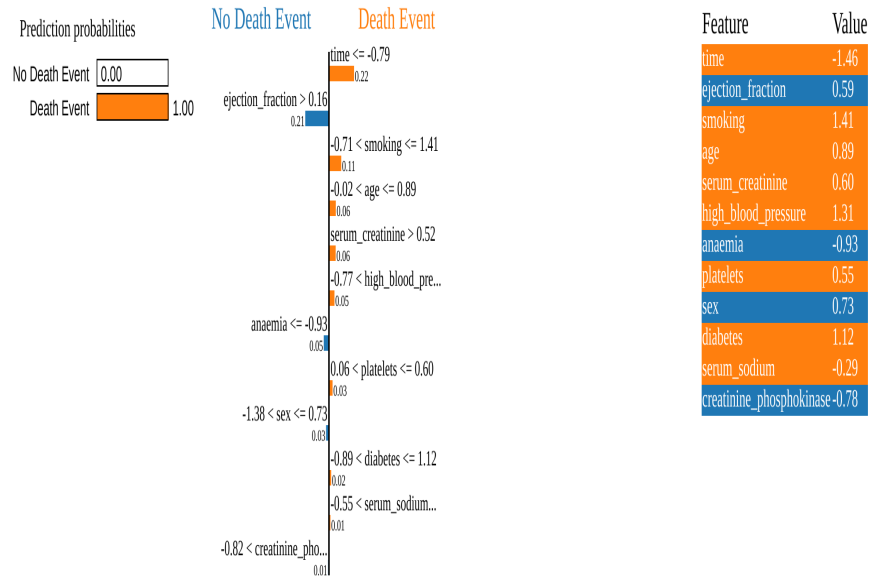


Figure 12: LIME summary for our KNN Model for a sample image

## Deep Learning Model: CNN-LSTM Architecture

For advanced cardiovascular disease prediction, a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks has been developed. The integration of these components aimed to capture complex non-linear relationships within the patient data for enhanced predictive accuracy.

## CNN-LSTM Model Hyperparameters

The performance of the CNN-LSTM deep learning model is highly dependent on the selection of its architectural hyperparameters. The key hyperparameters to be tuned for this model, along with their tested values and descriptions, are outlined in the table below:

Table 12: CNN-LSTM Model Hyperparameters and Their Descriptions

| Hyperparameter | Values  | Description                                                                                                                                                                                                                                                                      |
|----------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| seq_len        | 5, 6, 7 | Defines the length of the input sequence processed by the LSTM layer. This parameter directly influences the size of the flattened output from the LSTM, specifically affecting the number of nodes in the subsequent linear layer, calculated as $(seq\_len * n\_hidden) - 1$ . |
| n_hidden       | 4, 5, 6 | Corresponds to the <code>hidden_size</code> of the LSTM layer. It determines the number of hidden units (neurons) in the LSTM cells, impacting the model's capacity to learn complex patterns and representations.                                                               |
| n_layers       | 1, 2, 3 | Specifies the <code>num_layers</code> of the LSTM network. This controls the depth of the LSTM, indicating how many recurrent layers are stacked on top of each other. Deeper LSTMs can learn more abstract representations but require more data and computational resources.   |

### CNN-LSTM Model Training Results

The training and evaluation results for the CNN-LSTM model across different hyperparameter combinations are presented in Table 13. These results show the accuracy on the training, validation, and test sets for each configuration.

### Optimal CNN-LSTM Model Selection

From the comprehensive results of the hyperparameter tuning for the CNN-LSTM model, the optimal configuration is identified as:

- **Sequence Length (seq\_len) = 6**
- **Hidden Size (n\_hidden) = 4**
- **Number of Layers (n\_layers) = 3**

This particular model configuration achieved a training accuracy of **86.9%**, a validation accuracy of **82.8%**, and a test accuracy of **90.3%**. The nearly identical performance across the training, validation, and test sets indicates that this model has effectively learned the underlying patterns in the data without overfitting, demonstrating good generalization capabilities.

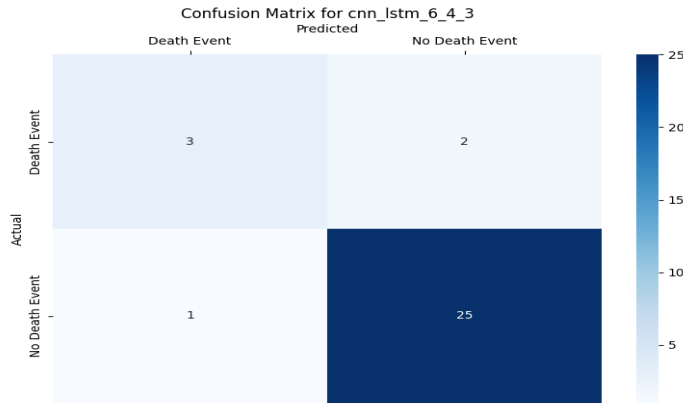


Figure 13: Confusion Matrix for the Optimal CNN+LSTM model

Table 13: CNN-LSTM Model Performance Across Hyperparameter Settings

| Seq. Length | N. Hidden | N. Layers | Acc. Train (%) | Acc. Val (%) | Acc. Test (%) |
|-------------|-----------|-----------|----------------|--------------|---------------|
| 5           | 4         | 1         | 92.8           | 82.8         | 80.6          |
| 5           | 4         | 2         | 94.2           | 82.8         | 80.6          |
| 5           | 4         | 3         | 91.3           | 82.8         | 77.4          |
| 5           | 5         | 1         | 94.2           | 82.8         | 74.2          |
| 5           | 5         | 2         | 92.8           | 82.8         | 77.4          |
| 5           | 5         | 3         | 92.0           | 86.2         | 83.9          |
| 5           | 6         | 1         | 97.1           | 82.8         | 77.4          |
| 5           | 6         | 2         | 95.7           | 82.8         | 77.4          |
| 5           | 6         | 3         | 89.9           | 86.2         | 71.0          |
| 6           | 4         | 1         | 79.6           | 72.4         | 83.9          |
| 6           | 4         | 2         | 89.1           | 69.0         | 87.1          |
| 6           | 4         | 3         | 86.9           | 82.8         | 90.3          |
| 6           | 5         | 1         | 93.4           | 58.6         | 83.9          |
| 6           | 5         | 2         | 92.7           | 75.9         | 83.9          |
| 6           | 5         | 3         | 87.6           | 65.5         | 83.9          |
| 6           | 6         | 1         | 94.2           | 79.3         | 90.3          |
| 6           | 6         | 2         | 89.1           | 69.0         | 90.3          |
| 6           | 6         | 3         | 88.3           | 69.0         | 83.9          |
| 7           | 4         | 1         | 93.4           | 86.2         | 60.0          |
| 7           | 4         | 2         | 78.1           | 82.8         | 66.7          |
| 7           | 4         | 3         | 89.8           | 86.2         | 73.3          |
| 7           | 5         | 1         | 94.2           | 82.8         | 70.0          |
| 7           | 5         | 2         | 89.8           | 89.7         | 66.7          |
| 7           | 5         | 3         | 81.8           | 89.7         | 80.0          |
| 7           | 6         | 1         | 96.4           | 82.8         | 80.0          |
| 7           | 6         | 2         | 89.1           | 89.7         | 70.0          |
| 7           | 6         | 3         | 78.1           | 82.8         | 66.7          |

### SHAP Interpretation for Optimal CNN-LSTM Model

The SHAP summary plot for the optimal CNN-LSTM model provides insights into the feature importance for its predictions. Given the sequential nature of the input to the CNN-LSTM, features are presented at different time steps (e.g., `t1_time`, `t2_time`). The plot shows the distribution of SHAP values for each feature across all instances, indicating their overall impact and direction of influence on the model's output.

Focusing on the most relevant features from this SHAP plot (considering only one 'time' feature to ensure diversity), we compare their overall impact with general global EDA insights:

Table 14: Comparison of Key Global EDA Insights and SHAP Explanation for Optimal CNN-LSTM Model

| Feature                     | EDA (Global Insight)                                                                   | SHAP (Overall Impact for CNN-LSTM Model)                                                                                                                     |
|-----------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t3_time                     | Generally, longer follow-up duration is associated with better outcomes (less risk).   | Most influential feature; higher values (●) tend to push the prediction towards increased risk of death, while lower values (●) push towards decreased risk. |
| t2_diabetes                 | Significant risk factor for cardiovascular disease.                                    | High impact; higher values (●) consistently increase the predicted risk of death.                                                                            |
| t3_creatinine_phosphokinase | Enzyme indicating muscle damage; higher levels can suggest heart issues.               | Moderate impact; higher values (●) can increase predicted risk, while lower values (●) can decrease it.                                                      |
| t2_ejection_fraction        | A key indicator of heart pumping efficiency; lower values typically imply higher risk. | Moderate impact; low values (●) increase predicted risk, while high values (●) decrease it.                                                                  |

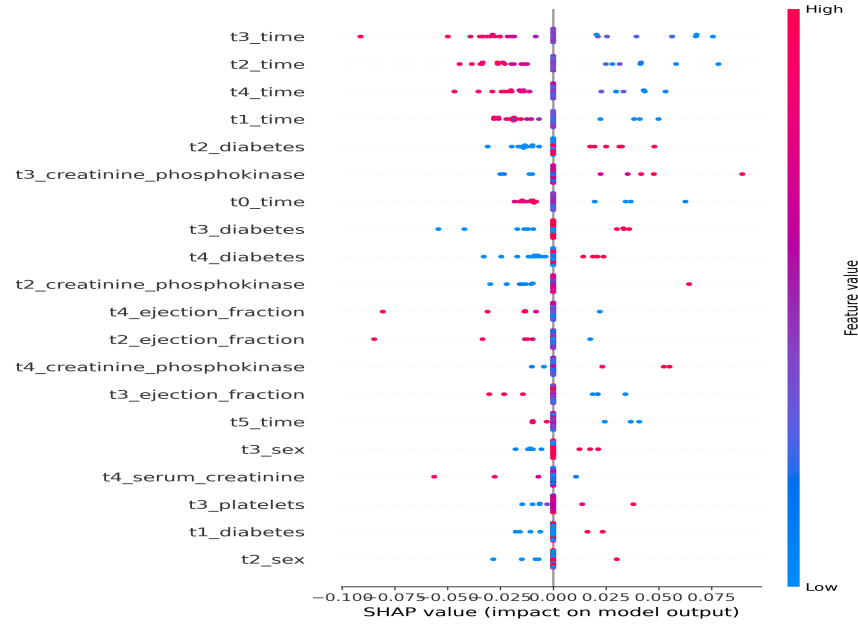


Figure 14: SHAP summary plot for our Optimal CNN+LSTM Model.

### LIME Interpretation for Optimal CNN-LSTM Model

The LIME explanation for a specific patient's prediction by the optimal CNN-LSTM model reveals that for this instance, the model predicts a "No Death Event" with 86% probability (and a 14% probability of "Death Event"). The local feature contributions driving this specific prediction are presented below, compared with general global insights from EDA, focusing on the most influential diverse features.

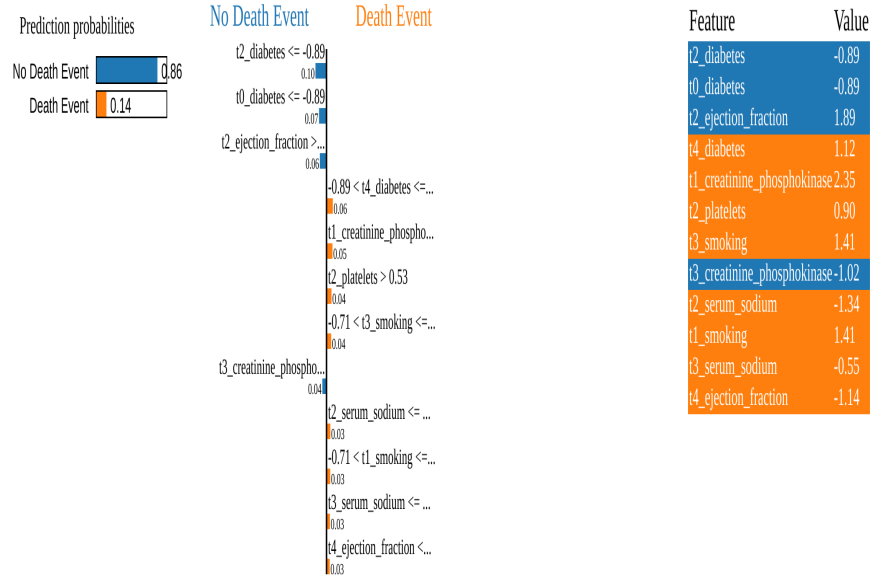


Figure 15: LIME summary plot for our Optimal CNN+LSTM Model.

Table 15: Comparison of Key Global EDA Insights and Local LIME Explanation for Optimal CNN-LSTM Model

| Feature                     | EDA (Global Insight)                                                                   | LIME (Local Explanation for this instance)                                 |
|-----------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| t2.diabetes                 | Significant risk factor for cardiovascular disease.                                    | Low value; notably pushes the prediction towards "No Death Event" (●).     |
| t2.ejection_fraction        | A key indicator of heart pumping efficiency; lower values typically imply higher risk. | High value; moderately pushes the prediction towards "No Death Event" (●). |
| t1.creatinine_phosphokinase | Enzyme indicating muscle damage; higher levels can suggest heart issues.               | High value; slightly pushes the prediction towards "Death Event" (●).      |
| t2.smoking                  | Major lifestyle risk factor for heart disease.                                         | High value; slightly pushes the prediction towards "No Death Event" (●).   |

## Conclusion

Our exploration of machine learning models for cardiovascular disease prediction yielded varied insights. **Gaussian Naive Bayes** and **Logistic Regression** established foundational benchmarks, with Logistic Regression identifying key features like **time** and **ejection\_fraction** via SHAP. The **Support Vector Classifier** showed robust training but some test set variability, highlighting **time** and **ejection\_fraction** as critical. **K-Nearest Neighbors** provided another baseline, with SHAP emphasizing **high\_blood\_pressure** and **time**. Finally, the **CNN-LSTM** deep learning model demonstrated strong generalization, with its explainability tools consistently pointing to **time**, **diabetes**, and **ejection\_fraction** as primary drivers.